

Solving right-angled triangles

Given any two parts, find the other three — the skill the whole chapter has been building towards.

LESSON 29–30

2 × 40 MIN

GEOMETRY 8, ТЕМЫ 25–26

STAGE 9 · BEYOND

LESSON CLOCK

5'

12'

8'

13'

2'

Warm-up	5 min
Explanation	12 min
Interactive	8 min
Practice	13 min
Homework	2 min

LEARNING OBJECTIVES

- Solve a right triangle given two sides.
- Solve a right triangle given one side and one acute angle.
- Choose the ratio that uses the two parts you know.
- Solve practical problems of height, distance and slope.

TEXTBOOK

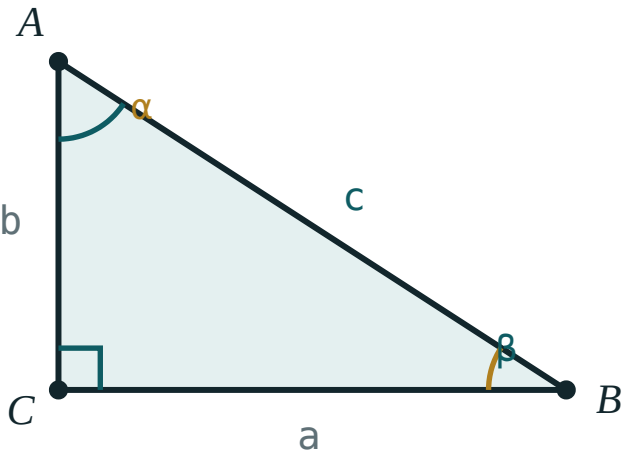
National	Темы 25–26, pp. 58–61
Cambridge	Extension beyond Stage 9
Workbook	—

01

Explanation

What “solving” means

A right triangle has six parts: three sides and three angles. One angle is already known (90°). Given any **two more** — as long as at least one is a side — the other three follow.



legs a , b · hypotenuse c

Six parts, one of them always 90° . Two more are enough to fix everything.

CHOOSING THE RATIO

Write down what you **know** and what you **want**, then pick the ratio containing exactly those two sides:

- opposite and hypotenuse $\rightarrow \sin$
- adjacent and hypotenuse $\rightarrow \cos$
- opposite and adjacent $\rightarrow \tan$ (or $\cot$)

Two sides and no angle? Use Pythagoras first, or a ratio to get the angle.

The two standard cases

Case 1 — two sides given. Find the third by Pythagoras, then an angle from a ratio, then the last angle by subtracting from 90° .

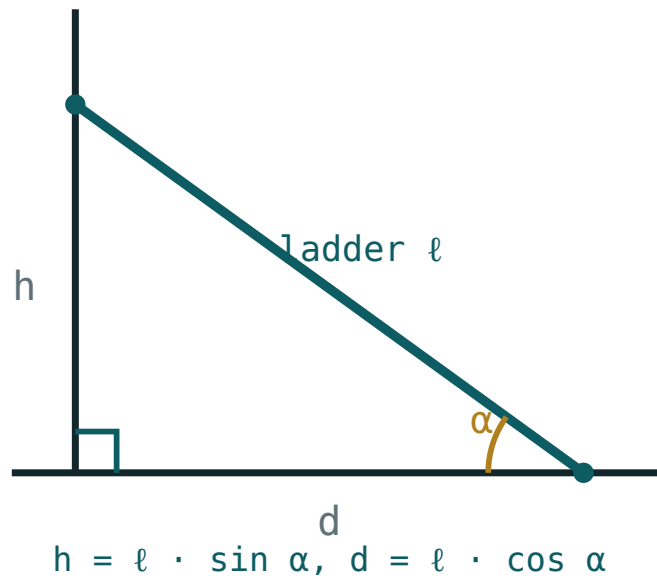
Case 2 — one side and one acute angle given. The other acute angle is $90^\circ - \alpha$ immediately. Then use the ratios to build the two missing sides:

$$a = c \cdot \sin \alpha \quad b = c \cdot \cos \alpha \quad a = b \cdot \tan \alpha$$

MULTIPLY OR DIVIDE?

Going from the hypotenuse to a leg you **multiply** by a ratio (the answer gets smaller). Going from a leg to the hypotenuse you **divide** (the answer gets bigger). If your hypotenuse comes out shorter than a leg, you divided the wrong way.

Practical problems



Every height-and-distance problem is this picture with different labels.

- **Angle of elevation** — measured *up* from the horizontal to the line of sight.
- **Angle of depression** — measured *down* from the horizontal.

They are alternate angles between two horizontals, so the angle of depression from the top of a tower equals the angle of elevation from the ground point. Draw the horizontal line at the observer's eye before anything else — most mistakes in these problems are mistakes in the diagram, not the trigonometry.

02 Worked examples**EXAMPLE 1** Solve the right triangle with $c = 20$ and $\alpha = 35^\circ$.

① $\beta = 90^\circ - 35^\circ = 55^\circ$

The other acute angle.

② $a = c \cdot \sin \alpha = 20 \cdot 0.5736 \approx 11.5$

Opposite and hypotenuse $\rightarrow$ sine.

③ $b = c \cdot \cos \alpha = 20 \cdot 0.8192 \approx 16.4$

Adjacent and hypotenuse $\rightarrow$ cosine.

④ check: $11.5^2 + 16.4^2 \approx 400 \checkmark$

Pythagoras confirms it.

ANSWER

$$\beta = 55^\circ, a \approx 11.5, b \approx 16.4$$

EXAMPLE 2 *Solve the right triangle with legs $a = 5$ and $b = 12$.*

$$\textcircled{1} \quad c = \sqrt{25 + 144} = 13$$

Pythagoras.

$$\textcircled{2} \quad \tan \alpha = \frac{5}{12} \approx 0.4167$$

Two legs $\rightarrow$ tangent.

$$\textcircled{3} \quad \alpha \approx 22.6^\circ$$

Inverse tangent.

$$\textcircled{4} \quad \beta \approx 67.4^\circ$$

$$90^\circ - 22.6^\circ$$

ANSWER

$$c = 13, \alpha \approx 22.6^\circ, \beta \approx 67.4^\circ$$

EXAMPLE
3**From a point 40 m from the foot of a tower the angle of elevation of its top is 38° . Find the height.**

- ① Draw the tower, the ground and the line of sight.

The tower is opposite the 38° angle.

② $\tan 38^\circ = \frac{h}{40}$

Opposite and adjacent $\rightarrow$ tangent.

③ $h = 40 \cdot \tan 38^\circ = 40 \cdot 0.7813$

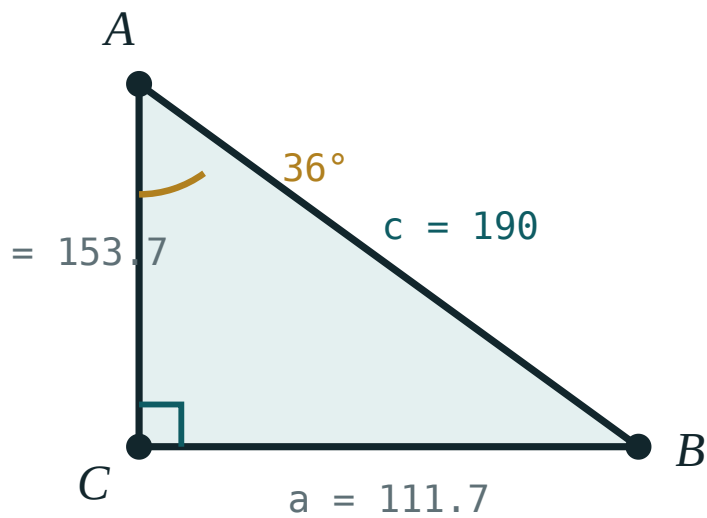
④ $h \approx 31.3 \text{ m}$

Sensible: less than the 40 m distance, since $38^\circ < 45^\circ$ ✓

ANSWER $\approx 31.3 \text{ m}$ **03 Interactive model**

Give the class two parts, let them predict the other three, then check with the model.

Solve the right triangle

 α 36° β 54° a (opposite) **111.7**b (adjacent) **153.7**c (hyp) **190** $\sin \alpha$ **0.59** $\cos \alpha$ **0.81** $\tan \alpha$ **0.73** $a^2 + b^2$ **36100** c^2 **36100****04 Terminology**

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Solve a triangle	Uchburchakni yechish	Решить треугольник
Given	Berilgan	Дано
Find	Topish kerak	Найти
Angle of elevation	Ko'tarilish burchagi	Угол возвышения
Angle of depression	Pasayish burchagi	Угол понижения
Slope	Qiyalik	Уклон
Height	Balandlik	Высота
Horizontal distance	Gorizontal masofa	Горизонтальное расстояние
Accuracy	Aniqlik	Точность

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Given the hypotenuse and the angle, the opposite leg is:

$$c \cdot \cos \alpha$$

$$c \cdot \sin \alpha$$

$$\frac{c}{\sin \alpha}$$

$$c \cdot \tan \alpha$$

Given both legs, to find an angle use:

$$\sin$$

$$\cos$$

$$\tan$$

Pythagoras

A leg is 6 and the angle opposite it is 30° . The hypotenuse is:

$$3$$

$$12$$

$$6\sqrt{3}$$

$$2$$

The angle of depression from a tower top equals:

the angle of elevation from the ground point

half of it

its complement

nothing in particular

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $c = 10, \alpha = 30^\circ$. Find a .

ANSWER 5

2. $c = 10, \alpha = 30^\circ$. Find β .

ANSWER 60°

3. $c = 8, \alpha = 45^\circ$. Find a .

ANSWER $4\sqrt{2} \approx 5.66$

4. $a = 3, b = 4$. Find c .

ANSWER 5

5. $a = 6, b = 6$. Find α .

ANSWER 45°

6. $c = 20, \alpha = 60^\circ$. Find b .

ANSWER 10

7. $\alpha = 25^\circ$. Find β .

ANSWER 65°

Medium · the standard question

7 PROBLEMS

1. $c = 20, \alpha = 35^\circ$. Solve the triangle.

ANSWER $\beta = 55^\circ, a \approx 11.5, b \approx 16.4$

2. $a = 5, b = 12$. Solve the triangle.

ANSWER $c = 13, \alpha \approx 22.6^\circ, \beta \approx 67.4^\circ$

3. $a = 7, c = 25$. Find b and α .

ANSWER $b = 24, \alpha \approx 16.3^\circ$

4. $b = 15, \alpha = 40^\circ$. Find a .

ANSWER $a = 15 \tan 40^\circ \approx 12.6$

5. $a = 9, \alpha = 30^\circ$. Find c .

ANSWER 18

6. From 40 m away the angle of elevation of a tower top is 38° . Find its height.

ANSWER ≈ 31.3 m

7. A ladder 6 m long makes 65° with the ground. How high does it reach?

ANSWER ≈ 5.4 m

Hard · stretch and reason

7 PROBLEMS

1. A ramp 15 m long rises 4 m. Find its angle of slope and its horizontal run.

ANSWER $\sin \alpha = 0.267$, $\alpha \approx 15.5^\circ$, run ≈ 14.5 m

2. From the top of a 50 m cliff the angle of depression of a boat is 22° . How far is the boat from the foot?

ANSWER $\frac{50}{\tan 22^\circ} \approx 123.8$ m

3. An isosceles triangle has equal sides 10 and apex angle 40° . Find its base.

ANSWER Half-base $10 \sin 20^\circ \approx 3.42$, so the base is ≈ 6.84 .

4. A rhombus has side 12 and an angle of 70° . Find both diagonals.

ANSWER $d_1 = 24 \sin 35^\circ \approx 13.8$, $d_2 = 24 \cos 35^\circ \approx 19.7$

5. Two people stand 100 m apart on opposite sides of a tower and see its top at 30° and 45° . Find the height.

ANSWER $h \cot 30^\circ + h \cot 45^\circ = 100$, so $h \approx 36.6$ m

6. A right triangle has area 24 and one leg 6. Find its angles.

ANSWER other leg 8, $\tan \alpha = \frac{8}{6}$, $\alpha \approx 53.1^\circ$, $\beta \approx 36.9^\circ$

7. A regular hexagon has side 6. Find the distance between two opposite sides.

ANSWER $2 \cdot 6 \sin 60^\circ = 6\sqrt{3} \approx 10.39$

67 Homework

Homework — 6 problems

Geometry 8, Темы 25–26, pp. 58–61. Draw and label a figure for every question.

- $c = 30$, $\alpha = 25^\circ$. Solve the triangle.
- $a = 8$, $b = 15$. Solve the triangle.

3. $a = 12$, $\alpha = 60^\circ$. Find c and b .
4. A ladder 8 m long makes 70° with the ground. How high does it reach?
5. From 60 m away the angle of elevation of a mast top is 42° . Find its height.
6. A ramp 20 m long rises 5 m. Find its angle of slope.